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Ryan; Kevin M. et al.

Hinged needle shield and needle assemblies

Abstract

A needle shield assembly comprises an elongate needle shield comprising a shield portion, a hub connection portion and a bridge portion including a living hinge having a thickness, the bridge portion joining the hub connection portion and the shield portion, the hub connection portion comprising a receiving ring having a diameter sized to frictionally receive the hub outer surface when the needle assembly is inserted through the receiving ring and a flange extending from the receiving ring defining a top planar surface and a bottom surface. The thickness of the flange divided by the thickness of the living hinge defines a ratio of from 1.1 to 6. The shield portion is configured to pivot from an open position in which the needle cannula is exposed, to a closed needle protecting position in which the distal end of the needle cannula is within the longitudinal opening of the shield.

Inventors: Ryan; Kevin M. (Whitehouse Station, NJ), Huenerfauth; Angela (Morristown, NJ), Pick; Thomas Anthony (Columbus, NE), Arazo; Ramon (Fraga Huesca, ES), Nogues Rubio; Miguel Luis (Torrente de Cinca, ES)

Applicant: Becton, Dickinson and Company (Franklin Lakes, NJ)

Family ID: 74068649

Assignee: Becton, Dickinson and Company (Franklin Lakes, NJ)

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Primary Examiner: Ahmed; Tasnim Mehjabin

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/017,263, filed on Sep. 10, 2020, now allowed, which claims priority to U.S. Provisional Application Nos. 62/900,147, filed on Sep. 13, 2019 and 62/901,603 filed on Sep. 17, 2019, the entire disclosures of which are hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to needle assemblies having safety features and more specifically to a needle assembly having a hinged needle shield assembly for use with a syringe.

BACKGROUND OF THE DISCLOSURE

(2) Accidental needle sticks with new unused needle can cause injury and render the needle unfit for further use. Accidental needle sticks with a used needle can transmit disease. As a result, most prior art needle assemblies have a needle shield. Some prior art needle shields define a rigid sleeve that can be manually telescoped over a needle cannula. This procedure requires a healthcare worker to hold the syringe barrel in one hand and the shield in the other. Some medical procedures require the application of pressure to the penetration site after the needle has been removed. Thus, healthcare workers are often unable to use both hands for shielding the needle cannula. In these situations, workers merely deposit the used medical implement on a nearby surface with the intention of shielding at a more convenient time. However, until the needle is shielded or properly disposed of, it presents a potential danger to other people.

(3) A needle shield which is hinged near the base of the needle has the advantage of allowing one-handed needle reshielding, providing the opportunity for reshielding by a medical practitioner, under most circumstances, immediately after use.

(4) Various means have been provided for locking a hinged needle shield in the closed, needle protecting, position. Deflectable members have been provided in the needle shield for engaging the needle upon shielding and preventing subsequent unshielding of the needle. Such members trap the needle within the needle shield. Locking has also been accomplished by locking engagement of the needle shield with structure near the base of the needle.

(5) There is still a need for an improved hinged needle shield that is cost effective and easy to manufacture.

SUMMARY OF THE DISCLOSURE

(6) A first embodiment pertains to a needle shield assembly for use with a needle assembly including a needle cannula having a proximal end, a distal end and a lumen therethrough, and a hub having an outer surface, an open proximal end including a cavity therein and a distal end attached to the proximal end of the needle cannula so that the lumen is in fluid communication with the cavity comprising an elongate needle shield comprising a shield portion, a hub connection portion and a bridge portion including a living hinge having a thickness, the bridge portion joining the hub connection portion and the shield portion, the hub connection portion comprising a receiving ring having a diameter sized to frictionally receive the hub outer surface when the needle assembly is inserted through the receiving ring and a flange extending from the receiving ring defining a top planar surface and a bottom surface, the flange having a thickness, the thickness of the flange divided by the thickness of the living hinge defining a ratio of from 1.1 to 6, the shield portion having two side walls defining a longitudinal opening and a back wall between the side walls defining a recess having an interior surface, the shield portion configured to pivot from an open position in which the needle cannula is exposed, to a closed needle protecting position in which the distal end of the needle cannula is within the longitudinal opening of the shield.

(7) A second embodiment pertains to a needle shield and needle assembly comprising a needle assembly including a needle cannula having a proximal end, a distal end and a lumen therethrough, and a hub having an outer surface, an open proximal end including a cavity therein and a distal end attached to the proximal end of the needle cannula so that the lumen is in fluid communication with the cavity; and an elongate needle shield comprising a shield portion, a hub connection portion and a bridge portion including a living hinge having a thickness, the bridge portion joining the hub connection portion and the shield portion, the hub connection portion comprising a receiving ring having a diameter sized to frictionally receive the hub outer surface when the needle assembly is inserted through the receiving ring and a flange extending from the receiving ring defining a top planar surface and a bottom surface, the flange having a thickness, the thickness of the flange divided by the thickness of the living hinge defining a ratio of from 1.1 to 6, the shield portion having two side walls defining a longitudinal opening and a back wall between the side walls defining a recess having an interior surface, the shield portion configured to pivot from an open position in which the needle cannula is exposed, to a closed needle-protecting position in which the distal end of the needle cannula is within the longitudinal opening of the shield.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is side elevation view of a needle assembly according to an embodiment of the present disclosure;
- (2) FIG. 2 is a top perspective view of the needle assembly shown in FIG. 1;
- (3) FIG. 3 a partial top perspective view of the needle assembly shown in FIG. 1;
- (4) FIG. 4 is a partial side elevation view of the needle assembly shown in FIG. 1;
- (5) FIG. 5 is an enlarged cross-sectional side elevation view of the distal end of the syringe assembly of FIG. 1;
- (6) FIG. 6A is a top partial perspective view of a portion of the needle assembly shown in FIG. 1;
- (7) FIG. 6B is a bottom partial enlarged perspective view of the receiving ring of the needle assembly shown in FIG. 1;
- (8) FIG. 6C is an enlarged perspective view of the hub of the needle assembly;
- (9) FIG. 7 is an exploded side elevation view of a needle assembly of the needle assembly shown in FIG. 1;
- (10) FIG. 8 is a perspective view of the needle assembly shown in FIG. 1 with the shield in a partially folded position; and
- (11) FIG. 9 is an exploded perspective view of a syringe and needle assembly.

DETAILED DESCRIPTION

(12) While this disclosure is satisfied by embodiments in many different forms, there are shown in the drawings and will herein be described in specific embodiments of the disclosure with the understanding that the present disclosure is to be considered exemplary of the principles and are not intended to limit the disclosure to the embodiments illustrated. The scope of the disclosure will be measured by the appended claims and their equivalents.

(13) Referring to FIGS. 1-9, a needle shield assembly **100** for use with a needle assembly **200** including a needle cannula **202** having a proximal end **203**, a distal end **205** and a lumen **206** therethrough. The needle assembly **200** further comprises a hub **208** having an outer surface **209**, an open proximal end **210** including a cavity **212** therein and a distal end **214** attached to the proximal end **203** of the needle cannula **202** so that the lumen **206** is in fluid communication with the cavity **212**.

(14) The needle shield assembly **100** comprises shield portion **102**, a hub connection portion **104** and a bridge portion **106** including a living hinge **108** having a thickness TH, the bridge portion

106 joining the hub connection portion **104** and the shield portion **102**. The hub connection portion **104** comprises a receiving ring **112** having a diameter D sized to frictionally receive the hub outer surface **209** when the needle assembly **200** is inserted through the receiving ring **112** and a flange **114** extending from the receiving ring **112** defining a top planar surface **116** and a bottom surface **118**. The distance between the top planar surface and the bottom surface **118** defines a flange thickness TF. In one or more embodiments, the thickness of the flange TF divided by the thickness of the living hinge TH defines a ratio of from 1.1 to 6. In one or more embodiments, the thickness of the flange TF divided by the thickness of the living hinge TH defines a ratio of from 1.2 to 6, 1.3 to 6, 1.4 to 6, 1.5 to 6, 1.6 to 6, 1.7 to 6, 1.8 to 6, 1.9 to 6, 2 to 6, 2.1 to 6, 2.2 to 6, 2.3 to 6, 2.4 to 6, 2.5 to 6, 2.6 to 6, 2.7 to 6, 2.8 to 6, 2.9 to 6, 3 to 6, 1.1 to 5, 1.2 to 5, 1.3 to 5, 1.4 to 5, 1.5 to 5, 1.6 to 5, 1.7 to 5, 1.8 to 5, 1.9 to 5, 2 to 5, 2.1 to 5, 2.2 to 5, 2.3 to 5, 2.4 to 5, 2.5 to 5, 2.6 to 5, 2.7 to 5, 2.8 to 5, 2.9 to 5, 3 to 5, 1.1 to 4, 1.2 to 4, 1.3 to 4, 1.4 to 4, 1.5 to 4, 1.6 to 4, 1.7 to 4, 1.8 to 4, 1.9 to 4, 2 to 4, 2.1 to 4, 2.2 to 4, 2.3 to 4, 2.4 to 4, 2.5 to 4, 2.6 to 4, 2.7 to 4, 2.8 to 4, 2.9 to 4, 3 to 4. (15) If the thickness of the living hinge TH is too great, the living hinge will not be flexible enough, and the shield will not be easily closed over the needle cannula. On the other hand, if the thickness of the living hinge TH is too small, the living hinge will be prone to breaking. In one or more embodiments, the ratio defined by the thickness of the flange TF divided by the thickness of the living hinge TH is within the above provided ranges. If the ratio defined by the thickness of the flange TF divided by the thickness of the living hinge TH is outside the defined ranges, the needle shield will be difficult to manufacture by injection molding because when the ratio is too low, insufficient material will flow to form the bridge portion **106** and the hub connection portion **104** of the needle shield assembly **100**. In one or more embodiments, in addition to being within the ratios provided above, the living hinge **108** has a thickness TH in a range of from 0.1 mm to 0.3 mm, for example 0.1 mm, 0.15 mm, 0.2 mm, 0.25 mm or 0.3 mm) and the flange TF **114** has a thickness in a range of from 0.4 mm to 0.8 mm (for example 0.4 mm, 0.45 mm, 0.5 mm, 0.55 mm, 0.60 mm, 0.65 mm, 0.70 mm, 0.75 mm or 0.80 mm). In addition, if the thickness of the flange **114** is too small, the flange **114** will have insufficient structural integrity to withstand insertion of the hub **208** through the receiving ring **112**. However, if the thickness of the flange TF is outside the upper end of the ranges provided and not within the ratios provided above, it will be difficult to form the part by injection molding.

(16) The shield portion **102** has two side walls, a first side wall **120** and a second sidewall **122**, defining a longitudinal opening **124** and a back wall **126** between the side walls **120**, **122** defining a recess **127** having an interior surface **128**. The shield portion **102** is configured to pivot from an open position in which the needle cannula is exposed (see e.g., FIG. 1 and FIG. 7) towards a closed protected position **180** (see FIG. 8) in which the distal end **205** of the needle cannula **202** is protected within the longitudinal opening **124** of the shield as shown in FIG. 8. It will be appreciated for the purposes of illustration that in FIG. 8, the protected position **180** is not in the fully closed position, but in a nearly closed position.

(17) In one or more embodiments, the receiving ring **112** further comprises a cylindrical wall extension **130** which extends below the bottom surface **118** of the flange **114** to provide a cylindrical receiving structure **130** for the hub. In one or more embodiments, the needle shield assembly comprises a plurality of ribs **132**, **134** extending from the bottom surface **118** and radially extending from the cylindrical wall extension **130**. In some embodiments, the plurality of ribs comprises from 3 to 8 ribs. In some embodiments, the plurality of ribs **132**, **134** comprises from 3 to 5 ribs, for example 4 ribs. The ribs **132**, **134** provide additional structural integrity to the flange **114**, which also allows the thickness of the flange TF to be reduced.

(18) In some embodiments, the bridge portion **106** comprising a first span **142** spaced apart from a second span **144** to provide a gap **146** between the first span **142** and the second span **144**, and the living hinge **108** having a first portion **108a** joining the first span **142** to the shield portion **102** and a second portion **108b** joining the second span **144** to the shield portion **102**. In one or more

embodiments, the first span **142** and the second span **144** each comprise a width **WS**, and the gap **146** comprises a width **WG** such that the width of the gap **WG** is greater than the width **WS** of the first span and the width **WS** of the second span.

(19) In one or more embodiments, the needle shield assembly **100** further comprises a leg **150** comprising a bend **152** connecting the shield portion **102** and the flange **114**. As shown in FIG. 3 and FIG. 4, the leg **150** is disposed beneath the gap **146**. The leg provides additional stability to the living hinge **108**.

(20) In some embodiments, the shield portion **102** further comprises a locking member **160** including two locking projections, a first locking projection **162** and a second locking projection **164**, facing each other and projecting from the side walls **120**, **122** and configured to engage with a locking tab **170**. The locking tab **170** includes a first tab **172** and ramp **173** which cooperates with the first locking projection ramp **163** of the locking member **160** and a second tab **174** and ramp which cooperate with the second locking projection ramp **165** of the locking member **160**. The first tab **172** and the second tab **174** of the locking tab **170** respectively push the first locking projection **162** and second locking projection **164** outwards. The first locking projection ramp **163** engages with the ramp **173** of the first tab **172** and the second locking projection ramp **165** engages with the ramp **173** of the second tab **174**. After the first locking projection ramp **163** and the second locking projection ramp **165** are moved beyond the ramp **173** of the first tab **172** and the ramp **175** of the second tab **174**, the locking projections **162**, **164** snap locks to the locking tab **170** to secure the shield portion **102** in a protecting position **180**. In this position, when the shield portion **102** is in the protected position **180** over the needle cover **240**, the device is in the ready to use position. A medical practitioner can grip the shield portion **102** and rotate the shield portion **102** away from the needle cover **240**, for example, as in the position shown in FIG. 1. When the practitioner is ready to inject a patient, the needle cover **240** is removed to expose the needle cannula **202**. After injection of the patient, the shield portion is rotated so that the shield portion **102** is in a protected position **180** over the needle cannula **202**.

(21) In one or more embodiments, the shield portion **102**, the bridge portion **106**, the hub connection portion **104** and the living hinge **108** comprise a unitary structure integrally molded of thermoplastic material.

(22) Another aspect of the disclosure pertains to a needle shield and needle assembly comprising the needle assembly **100** shown and described herein including the needle cannula **202** having a proximal end **203**, a distal end **205** and a lumen **206** therethrough, and a hub **208** having an outer surface **209**, an open proximal end **210** including a cavity **212** therein and a distal end **214** attached to the proximal end of the needle cannula **202** so that the lumen **206** is in fluid communication with the cavity **212**. The needle shield and needle assembly comprise the elongate needle shield **100** comprising a shield portion **102**, a hub connection portion **104** and a bridge portion **106** including a living hinge **108** having a thickness **TH**, the bridge portion **106** joining the hub connection **104** portion and the shield portion **102**. The hub connection portion comprises a receiving ring **112** having a diameter **D** sized to frictionally receive the hub outer surface **209** when the needle assembly **200** is inserted through the receiving ring **112** and a flange **114** extending from the receiving ring **112** defining a top planar surface **116** and a bottom surface **118**. The flange **114** has a thickness **TF**, the thickness of the flange **TF** divided by the thickness of the living hinge **TH** defining a ratio of from 1.1 to 6.

(23) In one or more embodiments, the thickness of the flange **TF** divided by the thickness of the living hinge **TH** defines a ratio of from 1.2 to 6, 1.3 to 6, 1.4 to 6, 1.5 to 6, 1.6 to 6, 1.7 to 6, 1.8 to 6, 1.9 to 6, 2 to 6, 2.1 to 6, 2.2 to 6, 2.3 to 6, 2.4 to 6, 2.5 to 6, 2.6 to 6, 2.7 to 6, 2.8 to 6, 2.9 to 6, 3 to 6, 1.1 to 5, 1.2 to 5, 1.3 to 5, 1.4 to 5, 1.5 to 5, 1.6 to 5, 1.7 to 5, 1.8 to 5, 1.9 to 5, 2 to 5, 2.1 to 5, 2.2 to 5, 2.3 to 5, 2.4 to 5, 2.5 to 5, 2.6 to 5, 2.7 to 5, 2.8 to 5, 2.9 to 5, 3 to 5, 1.1 to 4, 1.2 to 4, 1.3 to 4, 1.4 to 4, 1.5 to 4, 1.6 to 4, 1.7 to 4, 1.8 to 4, 1.9 to 4, 2 to 4, 2.1 to 4, 2.2 to 4, 2.3 to 4, 2.4 to 4, 2.5 to 4, 2.6 to 4, 2.7 to 4, 2.8 to 4, 2.9 to 4, 3 to 4.

(24) If the thickness of the living hinge TH is too great, the living hinge will not be flexible enough, and the shield will not be easily closed over the needle cannula. On the other hand, if the thickness of the living hinge TH is too small, the living hinge will be prone to breaking. In one or more embodiments, the ratio defined by the thickness of the flange TF divided by the thickness of the living hinge TH is within the above provided ranges. If the ratio defined by the thickness of the flange TF divided by the thickness of the living hinge TH is outside the defined ranges, the needle shield will be difficult to manufacture by injection molding because when the ratio is too low, insufficient material will flow to form the bridge portion **106** and the hub connection portion **104** of the needle shield assembly **100**. In one or more embodiments, in addition to being within the ratios provided above, the living hinge **108** has a thickness TH in a range of from 0.1 mm to 0.3 mm, for example 0.1 mm, 0.15 mm, 0.2 mm, 0.25 mm or 0.3 mm) and the flange TF **114** has a thickness in a range of from 0.4 mm to 0.8 mm (for example 0.4 mm, 0.45 mm, 0.5 mm, 0.55 mm, 0.60 mm, 0.65 mm, 0.70 mm, 0.75 mm or 0.80 mm).

(25) The shield portion **102** has two side walls, a first side wall **120** and a second sidewall **122**, defining a longitudinal opening **124** and a back wall **126** between the side walls **120**, **122** defining a recess **127** having an interior surface **128**. The shield portion **102** is configured to pivot from an open position in which the needle cannula is exposed (see e.g., FIG. 1 and FIG. 7), to a closed protected position **180** in which the distal end **205** of the needle cannula **202** is protected within the longitudinal opening **124** of the shield.

(26) In one or more embodiments, the needle shield and needle assembly further comprises a plurality of radial projections **220** extending from the needle hub, the radial projections **220** spaced from the open proximal end **210** so that the hub **208** remains seated in the receiving ring **112** when the needle assembly **200** has been inserted through the receiving ring. As best shown in FIG. 6A, the radial projections **220** are engaged with the periphery of the receiving ring **112** to prevent the hub **208** from being moved through the flange. The radial projections **220** are configured to prevent the hub **208** from being moved in a proximal direction to be pushed through the receiving ring **112**.

(27) In an exemplary use of the disclosure, a practitioner unpacks needle shield assembly **100** from packaging in an open position **190**. The practitioner assembles and secures syringe **310** to hub **208** of needle assembly **200**. Once syringe **310** has been engaged with the hub **208**, the needle cover **240** is removed from the needle assembly **200** and injection occurs. The practitioner removes the needle cannula **202** from the patient. Either with or without the syringe **310** secured to the hub **208** of the needle assembly **200**, practitioner rotates the hub connection portion **104** about the bridge portion **106** to toggle the needle shield assembly **100** from the open position **190** to the protected position **180**. In the protected position **180**, the locking tab **170** and the locking member **160** are fully engaged, thus shielding the needle cannula **202** from accidental post-injection needle sticks.

(28) Referring now to FIG. 6B, the cylindrical wall extension which extends below the bottom surface **118** of the flange **114** and provides a cylindrical receiving structure **130** for the hub **208** comprises an inner surface **135** and splines **136**. FIG. 6C shows the hub as having complementary hub splines **224** which are configured to engage the spline **136** on the inner surface **135** of the cylindrical receiving structure **130**. In the embodiment shown, the splines **136** are hollow sections on the inner surface **135**, and the hub splines **224** are elongate radial protrusions that are complementary to and configured to seat within the splines **136**. Thus, the splines **136** are female splines and the hub splines **224** are male splines. It will be understood that this figuration could be reversed. In the embodiment shown, twenty-four splines **136** are shown on the inner surface **135** of the cylindrical receiving structure **130**. In this configuration, the splines are space apart by 15 degrees around the periphery of the cylindrical receiving structure **130**. It will be understood that the number of splines can be varied so that the splines are spaced apart by 10 degrees (36 splines), 15 degrees (24 splines) 20 degrees (18 splines), 30 degrees (12 splines), 40 degrees (9 splines), 45 degrees (8 splines) 60 degrees (6 splines), 72 degrees (5 splines) or 90 degrees (4 splines). There are four hub splines **224** shown in FIG. 6C, and this number can vary and include one, two, three,

four, five, six, seven, eight, nine or then hub splines **224**.

(29) The hub splines **224** and the splines **136** are configured to cooperate to allow rotation of the hub **208** with respect to the shield portion **102**. The hub splines **224** and the splines **136** in some embodiments are configured to create a tactile or perceptible click during each rotation as the hub splines **224** and the splines **136** engage and disengage each other as the hub **208** is rotated with respect to the shield portion.

(30) In one or more embodiments, toggling the needle shield assembly **100** from open position **190** to protected position **180** may be completed in a variety of ways. During injection, the practitioner may be required to use one hand to press the injection site with a sanity cloth or pad during or after injection. The practitioner can remove the needle cannula **202** from the skin and single-handedly toggle the needle shield assembly **100** by rotating the syringe, thus using the weight of the shield portion **102** to fold over the hub connection portion **104** and fully engage the protected position **180**. Additionally, a practitioner may use a flat surface like a patient table, for example, to rotate the shield portion **102** about the bridge portion **106** to engage the protected position **180**.

(31) In one or more embodiments, a needle cannula clip (not shown) may be disposed within the longitudinal opening **124** of the shield portion **102** configured to receive the needle cannula **202** to increase shielding while the needle shield assembly **100** is in the protected position **180**.

(32) In one or more embodiments, after an injection is performed, a practitioner can remove the needle assembly **200** from the receiving ring **112** of the hub connection portion **104** while the needle shield assembly **100** is engaged in the protected position **180**.

(33) In one or more embodiments, the hub **208** is rotatable within the receiving ring **112**. The outside diameter of the hub **208** at the complementary hub splines **224** is configured so that there is a friction fit between the hub **208** and the receiving ring **112**, allowing rotation of the hub **208** to be set by the practitioner, but not too loose to allow free rotation. This configuration permits a proper or a preferred orientation of a bevel on the needle cannula **202** to be varied depending on a particular medical procedure. For example, subcutaneous, intravenous or intradermal types of injections typically require the needle to be inserted at various angles in relation to the patient's skin (e.g., 45 degrees, 25 degrees, 10 degrees respectively), with the needle bevel typically in the up position (facing away from the skin surface) and the bevel visible to the clinician. As such, it is advantageous for the hub **208** to be rotated to a position such that the needle bevel is not obstructing the clinician's view of the needle bevel. In intermuscular injection, the bevel is typically 90 degrees to the patient's skin. As such, the needle bevel rotational orientation in relation to the safety needle cover does not matter because the clinician does not need to see the bevel orientation.

(34) In one or more embodiments, the open proximal end **210** of the hub **208** comprises lugs **222** that are configured to be engaged with a threaded collar **315** of a syringe **310** having a distal end **312** and a proximal end **313** so that the needle shield and needle assembly connectable to the syringe **310**. As shown in FIG. 9, the syringe **310** may further comprise a plunger **320** having a distal end **322** and a proximal end **323**. The distal end **322** may include a stopper **324** having a distal tip **325**. In use the hub **208** shown in FIG. 9 can be inserted in the receiving ring **112** of the needle shield assembly **100**, and the syringe **310** can be connected to the hub by engaging the lugs **222** with threaded collar **314** of the syringe **310**.

(35) The plunger **320** can be positioned in fluid-tight engagement with the inside surface of the barrel of the syringe for drawing fluid into and driving fluid out of the barrel chamber by movement of the stopper relative to the barrel. The elongate body portion extends outwardly from the open end of the barrel.

(36) Although the disclosure herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing

from the spirit and scope of the disclosure. Thus, it is intended that the present disclosure include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A needle shield assembly comprising: a shield portion configured to pivot from an open position in which a needle cannula of a needle assembly is exposed, to a closed needle protecting position in which at least a distal end of the needle cannula is received within the shield portion; a hub connection portion; and a bridge portion including a living hinge, the bridge portion joining the hub connection portion and the shield portion, the hub connection portion comprising: a receiving ring having a diameter sized to frictionally receive a hub of a needle assembly when the needle assembly is received by the receiving ring, the receiving ring further comprising the cylindrical wall extension which extends below the bottom surface of the flange, and a flange extending from the receiving ring defining a top planar surface and a bottom surface, the receiving ring is rotatable relative to the hub, the cylindrical wall extension provides a cylindrical receiving structure having an inner surface with splines that are configured to cooperate with hub splines on the hub of the needle assembly when the hub of the needle assembly is engaged with the cylindrical wall extension, wherein the splines are configured to engage and disengage with the hub splines as the hub is rotated with respect to the shield portion to a fixed position.
2. The needle shield assembly of claim 1, further comprising a plurality of ribs extending from the bottom surface and radially extending from the cylindrical wall extension.
3. The needle shield assembly of claim 1, the bridge portion comprising a first span spaced apart from a second span to provide a gap between the first span and the second span, and the living hinge having a first portion joining the first span to the shield portion and a second portion joining the second span to the shield portion.
4. The needle shield assembly of claim 3, the first span and the second span each comprise a width and the gap comprises a width such that the width of the gap is greater than the width of the first span and the width of the second span.
5. The needle shield assembly of claim 4, further comprising a leg comprising a bend connecting the shield portion and the flange.
6. The needle shield assembly of claim 1, the living hinge having a thickness in a range of from 0.1 mm to 0.3 mm and the flange having a thickness in a range of from 0.4 mm to 0.8 mm.
7. The needle shield assembly of claim 1, the shield portion further comprising a locking member including two locking projections facing each other and projecting from the side walls and configured to engage with a needle cover placed over the needle cannula when the shield in the closed position to a locking tab extending from the flange.
8. The needle shield assembly of claim 1, the shield portion, the bridge portion, the hub connection portion and the living hinge comprising a unitary structure integrally molded of thermoplastic material.
9. A needle shield and needle assembly comprising: a needle assembly including a needle cannula having a proximal end, a distal end and a lumen therethrough, and a hub having an outer surface further comprising a plurality of radial projections and a plurality of hub splines extending radially from the hub, an open proximal end spaced from the radial projections and the hub splines including a cavity therein and a distal end attached to the proximal end of the needle cannula so that the lumen is in fluid communication with the cavity; and an elongate needle shield comprising a shield portion, a hub connection portion and a bridge portion including a living hinge having a thickness, the bridge portion joining the hub connection portion and the shield portion, the hub connection portion comprising a receiving ring having a diameter sized to frictionally receive the hub outer surface when the needle assembly is inserted through the receiving ring and a flange extending from the receiving ring defining a top planar surface and a bottom surface, the hub being

rotatable within the receiving ring, an outside diameter of the hub at the hub splines is configured to allow a friction fit between the hub and the receiving ring, allowing rotation of the hub to be set to a fixed position, the flange having a thickness, the thickness of the flange divided by the thickness of the living hinge defining a ratio of from 1.1 to 6, the shield portion having two side walls defining a longitudinal opening and a back wall between the side walls defining a recess having an interior surface, the shield portion configured to pivot from an open position in which the needle cannula is exposed, to a closed needle-protecting position in which the distal end of the needle cannula is within the longitudinal opening of the shield.

10. The needle shield and needle assembly of claim 9, the receiving ring further comprising a cylindrical wall extension which extends below the bottom surface of the flange to provide a cylindrical receiving structure.

11. The needle shield and needle assembly of claim 10, further comprising a plurality of ribs extending from the bottom surface and radially extending from the cylindrical wall extension.

12. The needle shield and needle assembly of claim 9, the bridge portion comprising a first span spaced apart from a second span to provide a gap between the first span and the second span, and the living hinge having a first portion joining the first span to the shield portion and a second portion joining the second span to the shield portion.

13. The needle shield and needle assembly of claim 12, the first span and the second span each comprise a width and the gap comprises a width such that the width of the gap is greater than the width of the first span and the width of the second span.

14. The needle shield and needle assembly of claim 13, further comprising a leg comprising a bend connecting the shield portion and the flange.

15. The needle shield and needle assembly of claim 9, the shield portion further comprising a locking member including two locking projections facing each other and projecting from the side walls and configured to engage with a needle cover placed over the needle cannula when the shield is in the closed position to a locking tab extending from the flange.

16. The needle shield and needle assembly of claim 9, the radial projections spaced from proximal end so that the hub remains seated in the receiving ring.

17. The needle shield and needle assembly of claim 16, the proximal end of the hub comprising lugs that are configured to be engaged with a threaded collar of a syringe so that the needle shield and needle assembly connectable to the syringe.

18. The needle shield and needle assembly of claim 9, wherein the splines are hollow sections on the inner surface, and the hub splines are elongate radial protrusions that are complementary to and configured to seat within the splines.

19. The needle shield and needle assembly of claim 18, wherein the hub splines and the splines are configured to cooperate to allow rotation of the hub with respect to the shield portion.
